

REVIEW PROTOCOL

*Requirements Specification – Seminar
Organization*

Course: Software Engineering
Polytechnic University of Tirana
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Review Team Information

Documents reviewed:	Requirements Specification v2.4.1 (Sections 3 & 7) Requirements Specification v3.1.5 (Sections 3 & 4) IEEE Std 1028-1997, Part 5
Team number:	6
Participants:	Erald Leka, Klajdi Oga, Sibora Muka, Erjona Haka, Kristiana Jorgo
Leader:	Erald Leka
Contributions by:	Erald Leka, Klajdi Oga, Sibora Muka, Erjona Haka, Kristiana Jorgo
Writer of protocol:	Erald Leka
Meetings (date/time/length):	3 meetings — 29 March 2026 (in-person, 1 hr and 10 min), 5 April 2026 (online, 45 min), 10 April 2026 (in-person, 1 hr and 20 min). Total: 3 hrs 15 min.

1. Summary Impression of Document v2.4.1 (Sections 3 and 7)

Section 3 – Product Functions

Section 3 provides a structured overview of elementary functions grouped into logical categories (Client Management, Seminar Management, Issuing Invoice, Queries). The use of unique identifiers (/F xx/) is a strength for traceability. However, the specification suffers from severe ambiguities, logical inconsistencies, factual errors and critical missing information. Many requirements are imprecise (vague terms like “useful”, “moral”), contradict other sections or earlier statements and lack necessary details for implementation.

Section 7 – Acceptance Test Cases

Section 7 is critically incomplete for a system with the scope defined in Section 3. Only 5 test cases (/T 10/ through /T 60/) are provided, leaving the vast majority of elementary functions with no acceptance test coverage at all. The existing test cases suffer from vague verification steps, undefined terms and illogical sequencing. Furthermore, no performance, boundary, negative or data persistence tests are present, despite measurable requirements being defined in Sections 5 and 6.

2. Problems of Document v2.4.1 Found in the Review Process (Sections 3 and 7)

2.1 Problems in Section 3

2.1.1 Inaccuracies

/F 10/ – Vague Terms

The statement “All relevant client data have to be delivered and maintained” is vague, as the term “relevant” is not defined. It is unclear which specific data attributes are required. There is a missing cross-reference to /D 10/ and no linkage to the data model defined in Section 4.

/F 30–36/

The requirement defines several verification steps, but they are not clearly specified or measurable. Checking whether the client is already registered (/F 31/) lacks a definition of how verification should be performed (e.g. by ID or email). No handling is defined if the desired seminar does not exist (/F 33/), if no free places are available (/F 34/) or if a duplicate booking is detected (/F 35/). The client's "payment moral" (/F 36/) is highly subjective, with no criteria or thresholds provided, making it untestable.

/F 32/ – Ambiguous Verification

The requirement “if the client data are still valid” is ambiguous, as no definition of “validity” is provided. The requirement does not specify whether validity is based on factors such as

data format (e.g. email structure), data completeness (e.g. missing fields) or time-based conditions (e.g. outdated address or expiry dates).

/F 40/

It does not specify the communication channel, the format of the invoice or what happens if delivery fails. The invoice should only be issued after the booking is confirmed, not simultaneously with the registration confirmation. This is reinforced by /F 350/, which specifies that the invoice is issued together with the certificate of seminar registration.

/F 50/ – Vague Actions

The requirement “recommend other seminars” is insufficiently specified, as no recommendation logic or criteria are defined. The term “useful” lacks a clear definition. There is no decision matrix or measurable criteria to determine what constitutes a relevant or appropriate recommendation.

/F 80–81/

It does not define how this check should be performed (e.g. based on an ID or booking record), nor does it specify any error handling in case the client is not found.

/F 82–83/

The expression “more than 4 weeks” is not precisely defined, as the exact time boundary is not specified. The term “80% charge fee” is also unclear, since it does not state 80% of what amount (e.g. total seminar price). The option to “deliver a substitute participant” is not defined in terms of process, leaving open questions such as who validates the substitute and at what point this substitution is accepted.

/F 90/

The communication channel is not specified (email, phone or letter), nor is the timing of notifications. It is unclear who triggers the notification and there is no error handling for unreachable participants due to invalid contact information.

/F 150/

The format and layout of the labels are not specified, leaving ambiguity in how the labels should appear. Authorization is also missing, as it is unclear who is allowed to produce the labels. There is no guidance on error handling if client data is incomplete, such as missing addresses or postal codes.

/F 180/

The term “payment moral” is subjective and not clearly defined, with no thresholds, metrics or criteria provided for evaluation. There is no description of how the assessment should be computed or measured.

/F 200/

There is ambiguity whether “seminars” refers to types (/D 50/) or instances (/D 40/). Key rules for creating, modifying or deleting seminars are missing, including limits on duration, participants and overlapping dates. Authorization and error handling are not specified and there is no guidance on how deletions affect existing bookings.

/F 260/

Does not specify how the system checks if the seminar took place (attendance list, lecturer confirmation, room booking). No rules for data validation. Vague about who performs the check (system or seminar manager). Lacks details on notifications or updates to seminar status. No information on handling discrepancies if the seminar did not occur.

/F 270/, /F 280/

Vague about format and delivery of the list (PDF, Excel, CSV). Missing error handling if data is incomplete. There is no specification of access control – it is not defined whether access is restricted to certain roles, nor is it stated who is authorized to generate the list.

/F 290/

Communication channel not defined. Does not specify communication method (email, printed copy, in-app notification). No rules for frequency or triggering events (before seminar, after seminar, after registration closes). No handling for missing or invalid email addresses. No access control (does the lecturer see only his/her seminars?).

/F 300/

Vague about format of certificate (digital, printed, signed). No information about batch generation or handling of special cases (cancellations, errors in participant data).

/F 350/

The use of “usually” is vague. It does not specify when or under what conditions the certificate and invoice are issued together. It does not define the sequence: should the invoice always follow the certificate or are they issued simultaneously? The requirement also lacks details about format, communication channel and timing, making it non-deterministic and hard to test.

2.1.2 Errors**Section 2 vs. /F 30/ – Contradiction on Client Types**

Section 2 defines that private clients book internal seminars and company clients book public seminars, whereas /F 30/ describes private clients booking seminars without distinguishing between internal and public types and no function enforces this rule. The overview contradicts Section 2 by stating that private clients book public seminars and company clients book internal seminars, creating an impossible mapping for implementation.

/F 60/ – Impossible Requirement

“Develop a new seminar which satisfies the wishes of the client” is not a software function but a business product development process, outside the system boundary. A software system cannot autonomously create seminars. This is a human/business decision, not a system function. It is also misplaced under Client Management instead of Seminar Management.

/F 84/, /F 220/ – Logical Error

There is a clear contradiction between the two requirements: /F 84/ states that if “Teachware” cancels a seminar, no invoice should be issued and the client is not charged, whereas /F 220/ states that if clients have already paid for a cancelled seminar, no reimbursement is provided.

These cannot both be true, as one implies no payment is required while the other implies payment without refund. This creates an illogical business rule that contradicts real-world expectations and legal obligations, with no justification or defined conditions.

/F 130/

Contains a critical referencing error: it states that “all verification steps like in the booking of private clients have to be taken into account (/F 20/ and the following points)”, but /F 20/ refers to modification and deletion of private clients rather than the actual booking verification steps, which are defined in /F 30–36/. The requirement also assumes that the system can automatically enforce the company delivering the participant name list, which mixes system responsibilities with human responsibilities and is therefore unrealistic. No function is specified for entering or modifying that name list.

/F 210/

It states that all consequences of a seminar removal are applied, yet /F 220/ specifies that clients who have already paid are not reimbursed. Since the overview excludes accounting, it is unclear who is responsible for handling refunds. Missing information includes automatic notifications to booked clients (only partially addressed in /F 220/), the timing of notifications and the responsibilities of the system versus human staff.

2.1.3 Missing Information

/F 20/

It does not specify who is authorized to perform these actions or any constraints, such as deleting a client with existing bookings or invoices. It also omits rules for handling related data, field validations, mandatory fields and audit/logging mechanisms needed for traceability.

/F 100/, /F 120/

Authorization rules and roles are not specified. Constraints are also undefined, such as whether a booking can be deleted if the seminar is full or if the client has already paid. The requirement fails to describe error handling for invalid or conflicting data, timing or concurrency issues, data validation rules and notifications to inform clients, staff or managers about changes.

/F 160/

It does not define how “interested partners” are determined (criteria or opt-in mechanisms), the mailing list format (e.g. CSV), error handling for invalid email addresses or the frequency and timing of generating the list.

/F 170/

Critical details are missing, such as which communication channel should be used by default, who is authorized to send circular letters and how to handle scheduling or logging to prevent duplicates or multiple sends. There is no guidance on error handling if some addresses are invalid.

/F 230/

No explicit link is defined between lecturers and seminar types (/D 65/). Important constraints are missing, such as ensuring unique IDs, maintaining data integrity and handling errors if a lecturer is deleted while assigned to a seminar. Does not specify who is authorized to perform these actions.

/F 250/

“Offer a seminar for a special semester”. No check that seminar type exists (/D 50/). Contradicts /T 60/ from Acceptance Test Cases. The specification is missing validation for minimum and maximum participants (/D 40/), lecturer availability (/D 65/) and room booking, which may lead to potential resource conflicts.

/F 360/

The requirement does not define which data sets are included, file format, file location or access rights/security. It is not specified how often the data is updated or if it is done automatically after each invoice. No error handling is mentioned if the file cannot be accessed or saved.

/F 400/, /F 410/

The requirement is vague about who can perform the query (client, company or staff). It does not specify how the system responds (format, channel, real-time update). It also omits error handling if no such seminar exists. Lacks clarity about time constraints: “next seminar on X” – “next” by what criteria (date, location)? “Attended” means completed or just booked?

/F 420/

Extremely vague and open-ended. It allows any query but gives no rules or constraints. Examples are given (top 10 companies by revenue, most-participated seminars), but no formal definition of query capability, data limits or performance expectations.

2.2 Problems in Section 7

2.2.1 Problems with Existing Test Cases

/T 10/ – Ambiguous and Incomplete`

The test mentions "client manager login" but no login function is defined anywhere in Section 3. There's no /F/ for authentication. Additionally "modify client data" is mentioned but the test doesn't say what gets modified or how to verify the change was actually saved.

/T 10/ – Illogical Test Sequence

The sequence “register new client, check out, new login, modify client data” is logically inconsistent. Checking out a client that was just registered and has not booked anything makes no sense as a test scenario.

/T 20/ – Missing Verification Steps

It states “cancel a seminar and inform clients” but does not specify how clients should be informed (email, letter), whether already paid fees are addressed or how to verify that all clients were actually reached.

/T 30/ – Vague and Open-Ended

Ends with “and others”, which is not acceptable in a test specification – a test case must list exactly what is being tested. The phrase “client data update necessary” is also undefined anywhere in the specification.

/T 50/ and /T 60/ – Mislabeled

Listed as “Acceptance Test Cases” but they actually describe data consistency rules, not test procedures. They state what must be true, not how to test it. These are constraints, not executable tests.

2.2.2 Missing Test Coverage**Cancellation Fee Logic (/F 82/, /F 83/, /F 84/)**

No test for the 100 EUR fixed fee (more than 4 weeks before), no test for the 80% fee (less than 4 weeks before) and no test for the case where Teachware cancels and no fee is charged. These are critical business rules involving money and have zero test coverage. The substitute participant option available in both /F 82/ and /F 83/ is also completely untested.

Company Bookings (/F 120/, /F 130/)

Company registration and internal seminar booking are completely untested, even though company bookings are a core part of the system.

Booking Verification Chain (/F 30/–/F 36/)

Booking has 6 sub-checks (already registered, data valid, seminar exists, free places, already booked, payment moral). /T 30/ vaguely touches some of these but does not test each one as a separate, verifiable case.

Client and Booking Management (/F 20/, /F 50/, /F 60/, /F 90/, /F 100/)

No test for booking management (registration, change, deletion) via /F 100/. No test for /F 50/ alternatives when booking fails. No test for /F 60/ (develop new seminar). No test for /F 90/ (inform all registered participants when Teachware cancels) – /T 20/ does not verify which participants were informed or that none were missed.

Mailing and Communication Functions (/F 150/, /F 160/, /F 170/)

Address labels (/F 150/), mailing lists (/F 160/) and circular letters to clients (/F 170/) are entirely untested.

Seminar Management (/F 180/, /F 200/, /F 210/, /F 220/, /F 230/, /F 250/, /F 260/)

Payment moral assessment (/F 180/), seminar development and deletion (/F 200/, /F 210/), the sensitive rule that no reimbursement is given for cancelled seminars (/F 220/), lecturer management (/F 230/), offering a seminar for a specific semester (/F 250/) and checking whether a seminar took place (/F 260/) are all completely untested.

Lecturer Communication and Certificates (/F 270/, /F 280/, /F 290/, /F 300/)

Circular letters to lecturers (/F 270/), participant lists (/F 280/), sending lists to lecturers (/F 290/) and certificate generation (/F 300/) have no test coverage whatsoever.

Invoice and Accounting (/F 350/, /F 360/)

Invoice issuing (/F 350/) and copies sent to accounting (/F 360/) have no acceptance tests.

Queries (/F 400/, /F 410/, /F 420/)

All three query functions are entirely untested.

2.2.3 Systematic Gaps

No Performance Tests

Section 5 defines measurable efficiency requirements: /E 10/ (response times: 15 sec for /F 250/ and /F 300/, 2 sec for all others), /E 20/ (max 100,000 participants, 10,000 seminars) and /E 30/ (5% late payers). None of these have corresponding acceptance tests.

No Data Persistence Tests

None of the /D/ data items (client, seminar, booking) are tested for being correctly saved, retrieved or maintained across sessions. The relationship defined in /D 65/ (lecturer presents a seminar type) is also never tested.

No Negative or Boundary Tests

No negative tests for invalid input (invalid emails, missing required fields, duplicate personal numbers). No boundary tests for /F 34/ (last available seat, overbooking by one) or for the participant limits defined in /D 40/ (participants_min and participants_max).

No Error Recovery Tests

No rollback or error recovery scenarios are tested. For example, if invoice sending (/F 40/) fails after the booking is already registered, it is not defined or tested whether the booking is rolled back. Section 7 contains no failure or mid-process error scenarios at all.

3. Relations Between Product Functions (v2.4.1) and Use Cases (v3.1.5)

/UC 10/

Informing (*from question to information*) covers the following elementary functions from v2.4.1:

- **/F 10/** Registration of private clients: All relevant client data (e.g. name, address, email) have to be delivered and maintained.
- **/F 20/** Modification and deletion of private clients.
- **/F 150/** Produce printed address labels for sending advertisements to all clients and companies, as well as for booking confirmation, invoices and other information.

/UC 20/

Booking (*from registration to booking*) covers the following elementary functions from v2.4.1:

- **/F 10/** Registration of private clients.
- **/F 20/** Modification and deletion of private clients.

- /F 30/ Booking a seminar by a client with verification:
- /F 31/ - if the client is already registered
- /F 32/ - if the client data are still valid
- /F 33/ - if the desired seminar really exists
- /F 34/ - if the seminar still has free places
- /F 35/ - if the client already booked the seminar
- /F 36/ - if the payment moral of the client is sufficient.
- /F 40/ Inform client about registration and send an invoice in case of positive verification results.
- /F 50/ In case of negative verification results: Do not book and inform client about the specific situation.
- /F 120/ Registration, modification and deletion of companies which send their associates to seminars.
- /F 180/ The accounting department registers all delayed payments to assess payment moral of clients.
- /F 350/ Issue certificate of seminar registration together with invoice.
- /F 360/ Copies of data sets considering invoices with be saved to a file, in order the accounting department can access it over network.

/UC 21/

Checking out (*from canceling to credit note*) covers the following elementary functions from v2.4.1:

- /F 20/ Modification and deletion of private clients.
- /F 80/ Client checks out a seminar booked before by him/her with verification:
- /F 81/ - if she/he was registered at all
- /F 82/ - if canceling happened more than 4 weeks before seminar, then 100 EUR fee
- /F 83/ - if canceling happened less than 4 weeks before seminar: then 80% charge fee
- /F 84/ - if “Teachware” canceled the seminar before: no invoice (no charged fee).

/UC 22/

Canceling (*from canceling to cancel notification*) covers the following elementary functions from v2.4.1:

- /F 90/ Inform all registered participants in case “Teachware” canceled the seminar.
- /F 210/ Remove seminars by “Teachware” with all consequences.
- /F 220/ In case there are already clients that already paid for the seminar canceled by “Teachware”, inform them that there is no reimbursement of those fees.
- /F 360/ Copies of data sets considering invoices will be saved to a file, in order the accounting department can access it over network (through a function).

/UC 23/

Booking company (*from registering to booking a company's internal seminar*) covers the following elementary functions from v2.4.1:

- /F 100/ Management of bookings: registration, change and deletion of bookings.
- /F 120/ Registration, modification and deletion of companies.
- /F 130/ A company can book an internal seminar with all verification steps.
- /F 350/ Issue certificate of seminar registration together with invoice.
- /F 360/ Copies of data sets considering invoices will be saved to a file, in order the accounting department can access it over network.

/UC 30/

Presenting seminar (*from participation to evaluation*) covers the following elementary functions from v2.4.1:

- /F 260/ Check if a seminar really took place after its announced final deadline has reached.
- /F 280/ Issue a list of all participants for the seminar X (seminar title, date from etc.).
- /F 290/ Send the participants list to lecturers and seminar managers.
- /F 300/ Issue a certificate for every participant (address, first name, second name, etc)

/UC 40/

Designing seminar (*from idea to a new seminar*) covers the following elementary functions from v2.4.1:

- /F 200/ Develop seminars: enter, change and delete seminars and seminar types.
- /F 230/ Register, change and delete lecturers including their data.
- /F 250/ Offer a seminar for a special semester with dates, times, lecturers, price.
- /F 420/ Other kinds of queries should be possible, e.g. seminar statistics.

/UC 50/

Acquiring lecturer (*from choosing to engaging*) covers the following elementary functions from v2.4.1:

- /F 230/ Register, change and delete lecturers including their data.

/UC 60/

Planning seminars (*from scheduling to reservation*) covers the following elementary functions from v2.4.1:

- /F 250/ Offer a seminar for a special semester with dates, times, lecturers, price.

/UC 70/

Participant list covers the following elementary functions from v2.4.1:

- /F 280/ Issue a list of all participants for the seminar X.
- /F 290/ Send the participants list to lecturers and seminar managers.

/UC 80/

Certificate covers the following elementary functions from v2.4.1:

- **/F 300/** Issue a certificate for every participant.

/UC 90/

Queries covers the following elementary functions from v2.4.1:

- **/F 400/** When will the next seminar on X take place?
- **/F 410/** Which associates of company Y have attended seminar X?
- **/F 420/** Other kinds of queries should be possible, e.g. biggest revenues per company during a financial year.

Elementary functions not covered by any use case:

- **/F 60/** In case the desired seminar does not exist, develop a new seminar which satisfies the wishes of the client.
- **/F 160/** In case of email, produce a mailing list of all interested partners.
- **/F 170/** A circular letter can be sent to all registered clients and companies, by mail or email.
- **/F 270/** A circular letter can be sent to all lecturers.

4. Comparison with IEEE Std 1028-1997, Part 5 – Technical Reviews

4.1 Introduction (Purpose)

Review protocol analyzes: **Requirements Specification**

Identifies:

- Inaccuracies
- Errors
- Missing information

In conclusion, Review Protocol fulfills the purpose of the technical review, because it focuses on the critical analysis of the document.

4.2 Responsibilities

In the Review Protocol are specified:

- Leader
- Team members
- Protocol writer

In conclusion, roles are defined but not formally aligned with IEEE role structure.

4.3 Input

In the Review Protocol are specified:

- Requirements Specification v2.4.1
- Requirements Specification v3.1.5
- IEEE Standard as a reference

In conclusion, the input is clear and correct according to the standard

4.4 Entry Criteria

In the Review Protocol:

- Explicit entry criteria are not mentioned
- Only meetings are mentioned

In conclusion, entry criteria are not explicitly defined, which deviates from IEEE requirements

4.5 Procedures

In the Review Protocol:

- **Planning**

-3 meetings have been held

-there is organization on time

- **Examination**

-detailed analysis of each function

- **Recording**

-the problems are clearly documented in categories:

- Inaccuracies
- Errors
- Missing information

- **Preparation**

-is not described how the reviewers were prepared and whether there was a checklist or guideline.

- **Follow-up**

-not mentioned whether the problems will be fixed and who will follow up actions after the review

In conclusion, some of procedures are structured according to the IEEE Standard while others do not match.

Exit criteria

In the Review Protocol there is a final report with:

- Analysis
- List of problems
- Comparisons

In conclusion, exit criteria are implicitly met and structured according to the standard but not declared in formal way.

4.6 Output

In the Review Protocol are included:

- Very detailed list of problems
- Identification of contradictions
- Gaps in test coverage
- Critical analysis

In conclusion, the output is very clear, complete and in accordance with IEEE Standard.

4.7 Conclusion

Review Protocol demonstrates a strong practical alignment with the objectives and outputs defined in IEEE Std 1028-1997 for technical reviews, particularly in its detailed defect identification, structured documentation, and critical evaluation of requirements.

However, the review protocol lacks formalization in several key procedural aspects required by the standard, including explicitly defined entry and exit criteria, clearly structured roles and responsibilities, and complete procedural phases such as preparation and follow-up.

Overall, the Review Protocol can be considered a partially compliant but highly effective practical implementation of a technical review, with strong emphasis on analysis and reporting, but limited adherence to the formal process framework defined by IEEE.

5. Remarks Concerning the Review Meeting

5.1 Contributions

Task distribution was as follows:

- Sibora conducted the review of Section 3 (Product Functions),
- Kristiana reviewed Section 7 (Acceptance Test Cases).
- Erald and Klajdi worked on the mapping of elementary functions to use cases.
- Erjona handled the comparison with IEEE Std 1028-1997 Part 5, and
- Erald additionally led the overall review process, supported the other team members throughout all tasks and was responsible for the final integration, writing and formatting of the complete protocol.

5.2 Preparation of Individual Participants

All team members read the required documents (Requirements Specification v2.4.1, v3.1.5 and IEEE Std 1028-1997 Part 5) individually before the first meeting on 29 March. Each member came prepared with their own observations and initial findings.

5.3 Main Points of Emphasis

The first meeting focused on task distribution. Several versions of the work division were proposed and discussed, and a final agreement was reached. The team discussed the content of the documents, analyzed the difficulty of each task and identified initial problems. Each member shared their perspective on how the work could best be approached.

The second meeting (online, 5 April) was held to follow up on progress. All members reported on the status of their tasks and shared difficulties encountered. Suggestions were made on how to structure the analysis and ensure the work maintained consistent quality throughout.

The third and final meeting (10 April) was dedicated to reviewing and correcting the last details in preparation for the final document. The protocol was read aloud by the team leader while all members followed along with their own copies, actively identifying any remaining gaps or inconsistencies. Suggestions were given to align the document as closely as possible with IEEE standards while preserving the originality of the team's analysis.

5.4 Organizational Notes

Coordinating schedules proved somewhat challenging, particularly for the online meeting, but all members managed to find suitable times. No major organizational problems were encountered.